

3 A person's immune response to viral infections involves T killer cells.

Cells infected with viruses present viral antigens to the T killer cells.

- (a) Peptide fragments bind to MHC molecules to form an antigen-MHC complex.
This takes place in the rough endoplasmic reticulum (rER).

The complex is then transported to the surface of the infected cell.

Describe how the antigen-MHC complex is transported from the rER onto the outside of the surface membrane of the infected cell.

(2)

- (b) An electron microscope with a resolution of $3.7 \text{ \AA}$ can be used to study antigen-MHC complexes.

$$1 \mu\text{m} = 1 \times 10^4 \text{ \AA}$$

- (i) Calculate the resolution of this electron microscope in nanometres.

(1)

Answer nm

- (ii) State what is meant by a resolution of $3.7 \text{ \AA}$

(1)

(c) Explain why antigens need to be presented to T killer cells in a person's immune response to viral infections.

(2)

.....

.....

.....

.....

.....

.....

(d) Explain how the action of T killer cells results in a person recovering from a viral infection.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 3 = 9 marks)